

The value of the acceleration due to gravity (g) in Cape Town

The value of the acceleration due to gravity (g) in Cape Town

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Introduction and Aim

All objects in the universe experience a gravitational force of attraction towards Earth, and the closer to the Earth's centre they are, the stronger the force is. This force creates an acceleration on the object, towards the centre of the Earth (generally taken as the $\hat{j}$ direction), and this acceleration will be a constant at a specific location on Earth and can be calculated. This is the acceleration due to gravity and the aim of this report is to calculate the value of this in Cape Town by observing the 1D aspect (the $\hat{j}$ direction) of the motion of objects which accelerate towards the Earth upon being released at some height

Method and Apparatus

Apparatus

- In order to create 6 videos that are used in the Tracker software, the following are needed:
 - o 6 different objects dropped, with a metre stick placed inside the video frame
 - The metre stick is used for calibration
 - o A phone to record the fall of the object
- A laptop is needed to analyse the data in the Tracker software

Methods

Videos were created by dropping objects from some height and recording their motion as they fall towards the Earth. The videos were opened in the Tracker software (using the  button). A calibration stick was added by clicking on the  icon and then navigating and clicking on the calibration stick option. Then the calibration stick was aligned with the metre stick placed in the video frame, with the length of the calibration stick being 1 metre. The same icon was clicked to stop showing the calibration stick. The axis was then aligned to the start position of the object, using the  button, and moving the grid onto the starting position of the object (and the grid can be unshown by clicking on the same icon). Then the slider was moved to just as the object was dropped, and this is marked as the start frame to slider (by right clicking the slider) and similarly the object just before it hits the ground is marked as start end frame to slider. The motion of the object is then tracked using the  Track menu and creating a new point mass. Then, starting from the start frame, some position was marked on the object by shift + clicking on that position of the object. This position was the same for every frame (such as the centre of ball or the bottle cap of a bottle) and the step forward  button was used to go to the next frame. The process of marking a position is repeated until all the frames are marked. Then the v_y vs t graph becomes available in the graph section to the right of the video for analysis as well as the (v_y, t) points which can be used in Excel to calculate the gradient of the line of best fit

Data and Results

Table 1: *Depicting the velocity values for the 6 objects at different times. This data is plotted in Figures 1→6*

A table showing the velocity (m/s) of different objects at different points of its path of travel, with each point having a time elapsed (s) since the first frame of the video associated with the ball dropping

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Time elapsed (s)	Velocity ($m \cdot s^{-1}$) for Squash ball	Velocity ($m \cdot s^{-1}$) for Tennis ball	Velocity ($m \cdot s^{-1}$) for Soccer ball	Velocity ($m \cdot s^{-1}$) for a bottle	Velocity ($m \cdot s^{-1}$) for bucket	Velocity ($m \cdot s^{-1}$) for tape roll
0,03333	-1,138	-1,529	-0,948	-0,932	-1,752	-1,160
0,06667	-1,474	-1,877	-1,284	-1,287	-2,097	-1,496
0,10000	-1,846	-2,164	-1,492	-1,716	-2,469	-1,835
0,13333	-2,218	-2,543	-1,835	-2,027	-2,761	-2,197
0,16667	-2,572	-2,906	-2,278	-2,353	-3,053	-2,557
0,20000	-2,961	-3,269	-2,599	-2,737	-3,425	-2,898
0,23333	-3,275	-3,663	-2,935	-3,063	-3,796	-3,279
0,26667	-3,592	-4,056	-3,302	-3,388	-4,088	-3,622
0,30000	-3,976	-4,298	-3,639	-3,744	-4,354	-3,999
0,33333	-4,305	-4,609	-3,791	-4,099	-4,619	-4,321
0,36667	-4,634	-4,904	-4,174	-4,409	-4,964	-4,592
0,40000	-4,969	-5,206	-4,632	-4,690	-5,203	-4,916
0,43333	-5,277	-5,509	-4,862	-4,986	-5,362	-5,187
0,46667	-5,609	-5,766	-5,091	-5,238	-5,575	-5,506
0,50000	-5,971	-6,084	-5,351	-5,549	-5,84	-5,752
0,53333	-6,352	-6,334	-5,703	-5,904	-6,239	-6,01
0,56667	-6,729	-6,705	-6,039	-6,170	-6,743	-6,328
0,60000	-7,076	-7,045	-6,329	-6,466	-6,982	-6,657
0,63333	-7,438	N/A	-6,452	-6,703	-7,141	-6,992
0,66667	-7,743	N/A	-6,834	N/A	N/A	-7,255

N/A means there were not enough frames in the motion to calculate this value

Example calculation: At each point marked, the Δr_y and the Δt is calculated (using $t_f - t_i$) and then

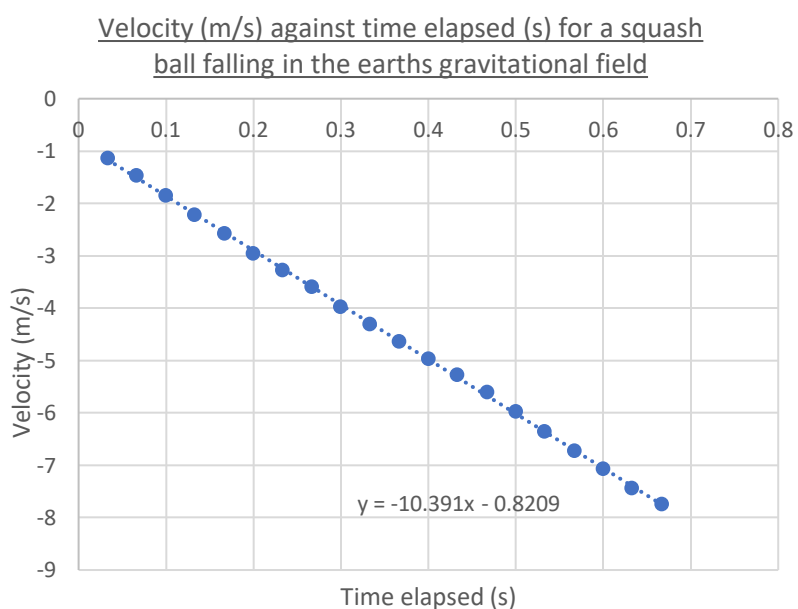
$$v_{y, \text{instant in middle of time interval}} \approx v_{y, \text{avg}} = \frac{\Delta r_y}{\Delta t}$$

Take the case: $t = 0.03333$. This is the first point marked and is in the middle of the interval: $0 \rightarrow 0.06667$, so $t_f = 0.06667$ and $t_i = 0$ so $\Delta t = 0.06667$

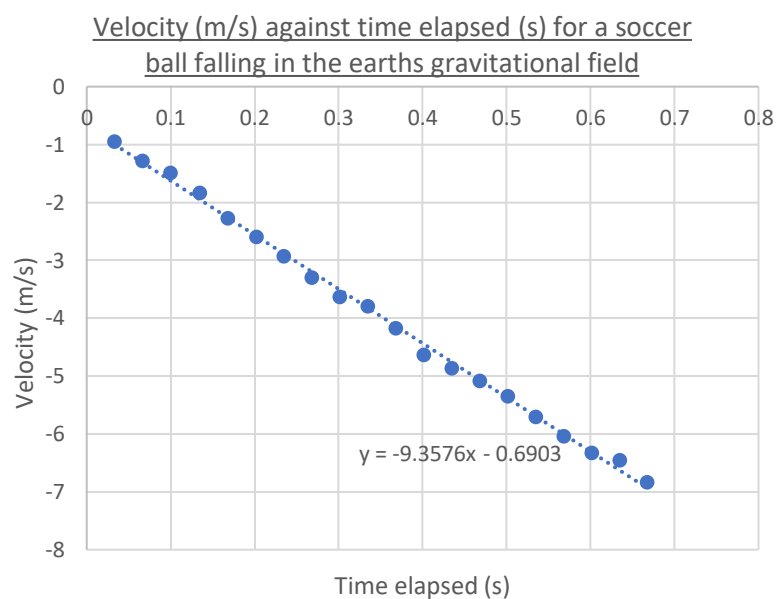
The program has calculated $r_{f,y} = -0.07650$. and $r_i = 0$ (the origin position), so $\Delta r_y = -0.07650$ then

$$v_{y, \text{instant in middle of time interval}} \approx v_{y, \text{avg}} = \frac{-0.07650}{0.06667} \approx -1.138 m \cdot s^{-1}$$

Graph 1: Calculating the acceleration for a squash ball by plotting the velocity for the squash ball in Table 1 against time

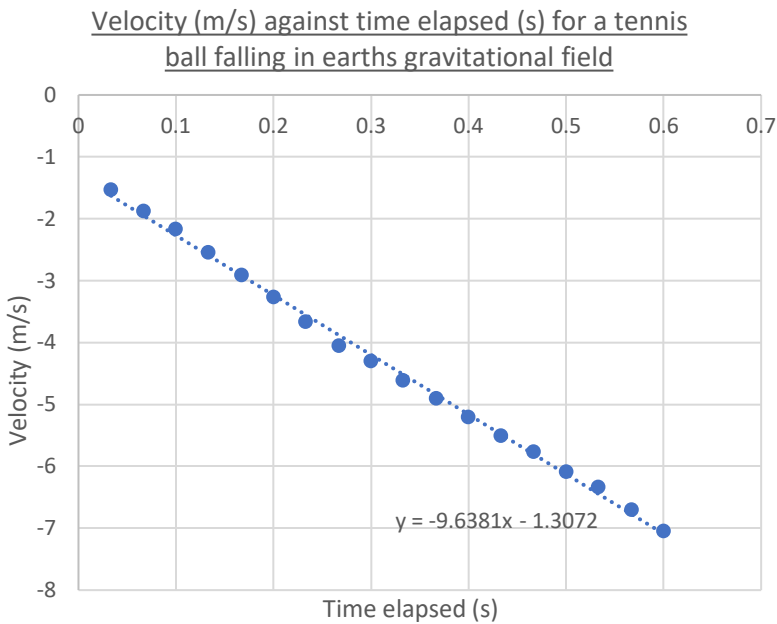


Graph 2: Calculating the acceleration for a soccer ball by plotting the velocity for the soccer ball in Table 1 against time

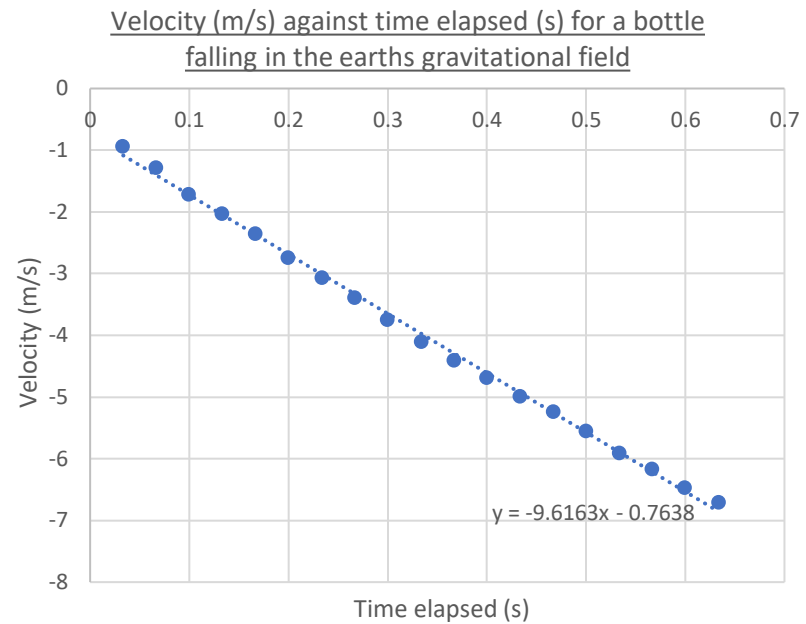


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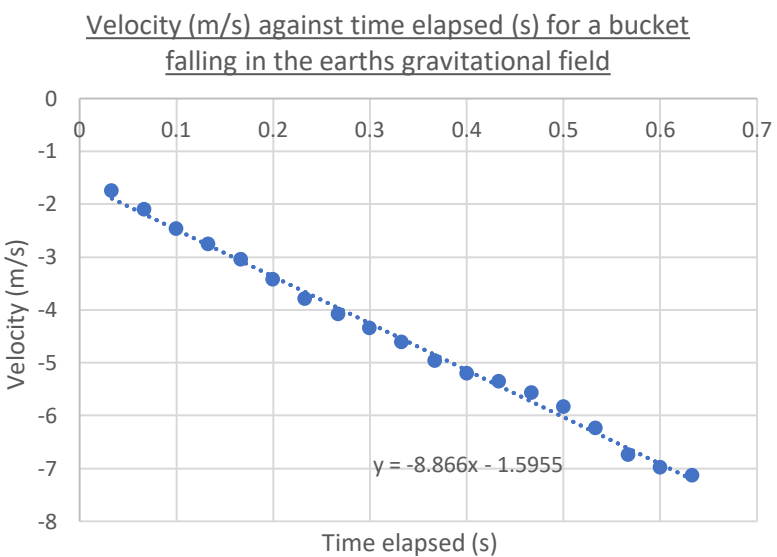
Graph 3: Calculating the acceleration for a tennis ball by plotting the velocity for the tennis ball in Table 1 against time



Graph 4: Calculating the acceleration for a bottle by plotting the velocity for the bottle in Table 1 against time



Graph 5: Calculating the acceleration for a bucket by plotting the velocity for the bucket in Table 1 against time



Graph 6: Calculating the acceleration for a tape roll by plotting the velocity for the tape roll in Table 1

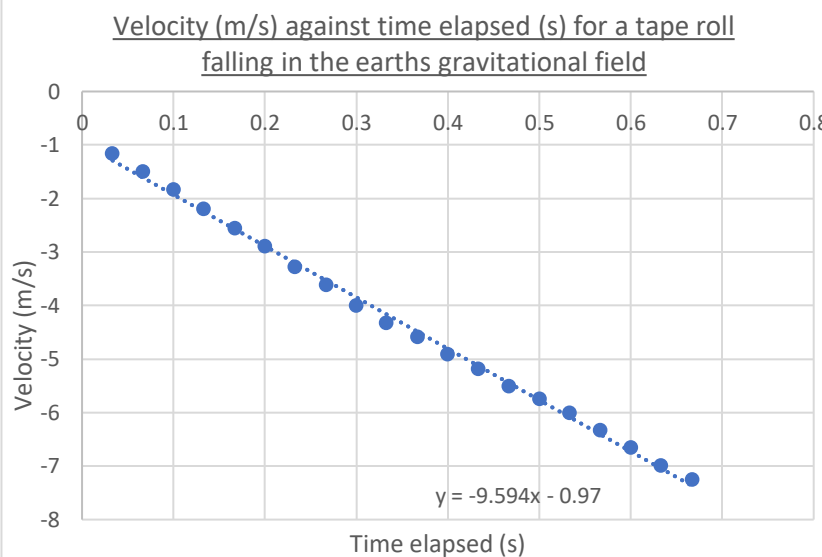


Table 2: This table shows the best estimate values for g from each of the Graphs 1 → 6 and their uncertainties (an example of the calculation is under Analysis)

Table showing the best estimate values for g and their uncertainty values:

Value for g ($m \cdot s^{-2}$)	Uncertainty ($m \cdot s^{-2}$)
-10,3910	0,03881
-9,3576	0,1181
-9,6381	0,1114
-9,6163	0,1083
-8,8660	0,1346
-9,5940	0,1109

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Analysis

Calculation of Uncertainty:

Example Calculation: This calculation will be done using the squash ball data
The uncertainty of the slope is calculated using:

$$u(m) = \sqrt{\frac{\sum d_i^2}{n \sum x_i^2 - (\sum x_i)^2} \left(\frac{n}{n-2}\right)}$$

where $d_i = y_i - (mx_i + c)$ and $(mx_i + c)$ is the line of best fit in the graphs above
Then the uncertainty for the data above is calculated with (using Graph 1):

$$u(m) = \sqrt{\frac{\sum (v_i - (-10.391x_i - 0.8209))^2}{20 \sum x_i^2 - (\sum x_i)^2} \left(\frac{20}{18}\right)}$$

The LINEST function on excel does this calculation and returns the uncertainty

Calculation of g

We have that the best approximation for g is then (using the values in Table 2):

$$g = \frac{-10.3910 - 9.3576 - 9.6381 - 9.6163 - 8.8660 - 9.5940}{6}$$

which gives:

$$g = -9.57717$$

$$\text{Then } u(g) = \frac{1}{6} \sqrt{0.03881^2 + 0.1181^2 + 0.1114^2 + 0.1083^2 + 0.1346^2 + 0.1109^2} = 0.04410$$

Finally:

$$g = -9.57717 \pm 0.04410$$

Discussion and Conclusion

From this it can be concluded that $g = -9.57717 \pm 0.04410$ in Cape Town (as shown under the analysis section). The aim was fulfilled because although 9.8 (the generally stated g for a frictionless world) is not within this range, the best estimate is slightly less than this which is what is expected in a scenario involving friction.

Sources of Uncertainty

- The limited frame rate and quality of a video affects the quality of the results.
- The judgement in marking the same spot on the object in the Tracker software (such as the centre of mass of a ball)
- The sensitivity of the Tracker software to calculate values with a limited precision (due to memory)

Recommendations

- Provide some indication on the objects that can be used to mark the position of the objects on the Tracker software, such as a red sticker at the middle of a ball. This will allow for greater precision, with less uncertainty associated with the judgement in marking the same spot on the object in the Tracker software
- Drop the object with minimal wind (if wind is present) and drop the object without any sort of spin so as to eliminate using some indication which is not the centre of mass which can be used in the Tracker software (such as a bottle lid)
- Film the videos with a greater frame rate and quality (for the same reasons as above)



PHY1004W: Assessment criteria for freefall practical report.

Peoplesoft ID: _____

Date of experiment: _____

Marked by: _____

Marking Guidelines

The person marking your report will make one of the following marks inside the box next to each criterion:

- ☒ is adequate (1 point),
- ☐ needs improvement (0.5 points), and
- ☒ is inadequate or missing (0 points)

It is important that you understand the mark that you receive and speak to someone from lab team about it.

Title, introduction and aim (3 marks):

- ☐ The report has a suitable title.
- ☐ Peoplesoft ID and date are included.
- ☐ The aim (purpose) of doing the experiment is clearly stated.

Method and apparatus (2 marks):

- ☐ All apparatus (including diagrams with suitable headings and labels) are described briefly.
- ☐ The method is written in a paragraph. It describes briefly and clearly significant steps in the procedure (including analysis of data).

Data and tables (4 marks):

- ☐ An appropriate number of falling objects are reported on.
- ☐ All useful results have been recorded correctly in tables. (Tables of position-time data should not be included because they are captured in the graphs.)
- ☐ Uncertainty analysis is correct for all measurements.

- ☐ All tables have suitable captions, and columns in tables have correct SI units.

Graphs (5 marks):

- ☐ The data have been linearised and are plotted correctly.
- ☐ Graphs have headings which state why they were plotted.
- ☐ Appropriate scales used.
- ☐ Axes are labeled correctly, including correct units.
- ☐ A straight line and a curve are appropriately "fitted" to the data.

Analysis (3 marks):

- ☐ The analysis is correct and easy to follow.
- ☐ Values of g are correctly obtained from graphs.
- ☐ Uncertainties for g have been estimated and are correct.

Discussion and conclusions (5 marks):

- ☐ All results (best approximation and standard uncertainty) obtained are clearly stated.
- ☐ There is some discussion on the validity of the results obtained for g .
- ☐ There is suitable discussion around factors that might have influenced measurements.
- ☐ There is suitable discussion around possible ways of improving the experiment.
- ☐ The conclusions are appropriate and are related to the aim of the experiment.

Overall impression (3 marks):

- ☐ The experimental details are understood and communicated briefly.
- ☐ The report is easy to follow. It is presented in a logical sequence.
- ☐ Grammar is adequate.